Exploring Mixtures and their Separation

Chapter 5 | Exploration, Grade 9 (NCERT) | Quick Revision Notes

5.1 Classification of Mixtures

- **Homogeneous mixture (solution):** uniform composition throughout; particles not visible. Eg. sugar in water, vinegar, soda, air, brass.
- **Heterogeneous mixture:** non-uniform; components visible and may settle. Eg. sand in water, oil and water, smoke, fog.
- Mixtures are also classified as **solution, suspension or colloid** (by particle size).

5.2 Solutions and Concentration

- **Solute** = substance that dissolves; **solvent** = substance that dissolves the solute (sugar in water: sugar is solute, water is solvent).
- **Concentration** = amount of solute dissolved in a given amount of solvent or solution. Right proportion matters (ORS, pesticide spray, medicines).
- Total mass of solution = mass of solute + mass of solvent.

Type	Formula	Used for	Example
Mass by mass (% m/m or w/w)	(Mass of solute / Mass of solution) × 100	Homogeneous mixtures, packaged foods, industry	10 g salt in 90 g water = 10% m/m
Mass by volume (% m/v)	(Mass of solute / Volume of solution) × 100	Medicines, laboratories	5 g glucose in 100 mL = 5% m/v; saline drip = 0.9% m/v
Volume by volume (% v/v)	(Volume of solute / Volume of solution) × 100	Two miscible liquids (perfume, vinegar)	1 mL pesticide in 100 mL = 1% v/v; vinegar 5% v/v

Solubility

- **Solubility** = maximum amount of solute that dissolves in a fixed quantity of solvent (100 g or 100 mL) at a given temperature.
- **Saturated solution** = cannot dissolve more solute at that temperature.
- Solid in liquid: solubility **generally increases** with temperature. Gas in liquid: solubility **generally decreases** with temperature.
- **Solubility curve** = graph of solubility (g per 100 g water) vs temperature. A steeper curve means solubility is more affected by temperature (compound 'B' in Fig. 5.6 rises more than 'A').

5.3 Separation of Homogeneous Mixtures

1. Crystallization

- Forming pure **crystals** (solids with particles in a regular geometric pattern) from a hot saturated solution on cooling. Based on **difference in solubility at different temperatures**.
- Eg. 287 g of 'B' in 100 g water at 60 °C; cooled to 40 °C, only 241 g stays dissolved, so 46 g separates as crystals.
- **Steps (copper sulfate):** make saturated solution (warm, add a drop of dilute H_2SO_4) → filter hot to remove insoluble impurities → cool **slowly** and undisturbed → filter crystals, rinse with cold water, dry.
- **Slow cooling gives larger, well-shaped crystals**; rapid cooling gives small, poorly formed ones.
- Uses: purification of solids, separating two soluble solids when one is in small amount, salt from seawater (seawater → saturated solution → salt crystals).
- **Evaporation vs crystallization:** evaporation removes all solvent (impurities may remain); crystallization gives pure crystals and does not dry up the solution.

2. Distillation

- Separates **miscible liquids** whose boiling points differ by **at least about 25 °C**; also recovers a solvent from a solution with dissolved solids.
- **Principle:** heat → liquid with lower boiling point vaporises → vapour cooled in a **condenser** → pure liquid (**distillate**) collected. Higher boiling liquid/solid stays in the flask.
- Eg. acetone (56 °C) and water (100 °C): acetone distils first.

- **Apparatus:** distillation flask, thermometer, water condenser (water enters at bottom inlet, leaves at top outlet), conical flask, burner, tripod, wire gauze.
- **Fractional distillation:** for liquids with boiling points differing by **less than 25 °C**; eg. crude petroleum into petroleum gas (LPG), petrol, kerosene, diesel, lubricating oil, bitumen.
- India: *Deg-Bhapka* method of perfume (ittar) making in Kannauj; ancient salt making (*panga* and *karkatch* salt).

3. Paper Chromatography

- Separates components (eg. dyes in black ink, pigments in spinach or petals) by **differences in their rates of movement** on paper, depending on interaction with solvent and paper. Word means 'writing with colour'.
- **Method:** draw a pencil line 2 cm from the bottom of the strip → put an ink spot on it → dip the lower end in solvent (water) with **solvent level below the spot** → solvent rises and carries the components different distances, forming separate coloured spots.
- If water does not work, use another solvent such as alcohol or a solvent mixture.

5.4 Separation of Heterogeneous Mixtures

1. Separating funnel (two immiscible liquids)

- Immiscible liquids (oil and water) form **two layers based on density**: the less dense liquid (mustard oil) floats on top; denser water is the lower layer.
- Open the stopcock slowly, drain the lower layer, close before the interface, discard the mixed portion, then collect the upper layer.

2. Sublimation

- **Sublimation:** solid changes directly to vapour on heating (below its melting point) without becoming liquid. **Deposition:** vapour cools and turns directly back into a solid.
- Separates a sublimable solid from a non-sublimable one: camphor, naphthalene or ammonium chloride from sand/salt. Camphor deposits on the inverted funnel wall; sand stays in the china dish.
- Dry ice (solid CO_2) also sublimes. **Sublimation is different from evaporation** as it skips the liquid state.

3. Suspensions

- Heterogeneous mixtures where insoluble solid particles are **visible, do not dissolve** and **settle on standing**. Eg. muddy water, sawdust in water, tea leaves in water.
- **Filtration** removes larger particles but fine particles may remain (water stays cloudy). Then use centrifugation or coagulation.

4. Centrifugation

- Spinning a mixture at high speed; **centrifugal force** pushes heavier particles outwards (to the bottom of the tube) while the lighter liquid stays on top.
- Uses: separating blood into red blood cells, platelets, white blood cells and plasma; chemical industries.
- **Paperfuge:** low-cost, hand-powered (toy-like) centrifuge, needs no electricity, helps detect malaria and anaemia in remote areas.

5. Coagulation

- A **coagulant** (eg. powdered alum/fitkari) makes fine suspended particles **clump together**; the bigger clumps settle by gravity (**sedimentation**) and are removed by decantation or filtration.
- Also: paneer from milk (lemon juice or vinegar coagulates milk proteins); sewage treatment (sedimentation, coagulation, filtration).

Colloids, Tyndall Effect and Alloys

Property	Solution	Suspension	Colloid
Nature	Homogeneous	Heterogeneous	Appears homogeneous (heterogeneous at microscopic level)
Particle size	Less than 1 nm	More than 1000 nm	1 to 1000 nm
Visibility	Not visible; transparent	Visible to naked eye	Not visible to naked eye
Filtration	Cannot be separated	Can be separated	Cannot be separated by ordinary filtration

Property	Solution	Suspension	Colloid
Settling	Does not settle	Settles on standing	Does not settle
Tyndall effect	No	Yes	Yes
Examples	Salt/sugar solution, vinegar, soda	Muddy water, sand in water, chalk in water	Milk, blood, tomato sauce, ice cream, butter, smoke, fog, clouds

- **Tyndall effect:** scattering of light by particles, making the path of a light beam visible (named after John Tyndall). Shown by colloids and suspensions, **not by true solutions**. Eg. sunbeam through a dusty room or leaves, floodlights in a stadium, smoky/hazy cities.
- **Dispersed phase** = solute-like particles in a colloid; **dispersion medium** = the component in which they are spread. In blood: dispersed phase is the blood cells (and other components), medium is plasma.
- **Emulsion:** colloid with both phases liquid. *Oil-in-water*: milk, vanishing cream. *Water-in-oil*: butter, cold cream, body lotion. An **emulsifying agent** (eg. milk proteins, soap) stabilises it.
- **Alloy:** homogeneous mixture of two or more metals, or a metal and a non-metal; cannot be separated by physical methods. Brass = about 80% Cu + 20% Zn; bronze = about 80% Cu + 20% Sn; stainless steel = iron + carbon, chromium, nickel, molybdenum.

Quick Guide: Which Method to Use?

Mixture	Method	Property used
Pure solid from its saturated solution	Crystallization	Solubility changes with temperature
Solvent from solution / miscible liquids (bp difference 25 °C or more)	Distillation	Different boiling points
Miscible liquids (bp difference less than 25 °C)	Fractional distillation	Small bp differences
Dyes, pigments, ink components	Paper chromatography	Different rates of movement
Oil and water	Separating funnel	Immiscible; different densities
Camphor/naphthalene and sand	Sublimation	One solid sublimes
Blood components, fine suspended solids	Centrifugation	Different densities (heavier moves outward)
Muddy water, fine clay particles	Coagulation (alum) + sedimentation, decantation, filtration	Particles clump and settle
Sand + salt + naphthalene	Sublimation (naphthalene) → dissolve in water and filter (sand) → evaporate/crystallize (salt)	Combine properties stepwise

Remember These Points

- Always give **units and % type** when expressing concentration; mass of solution = solute + solvent.
- Chromatography: **pencil** line (does not dissolve), solvent level **below** the spot.
- Distillation: condenser water enters from the **lower inlet**; thermometer bulb at the flask's side-arm.
- Slow cooling = bigger, purer crystals. Crystallization is not the same as evaporation.
- Solution: no Tyndall effect, no settling. Suspension: settles, filterable. Colloid: no settling, shows Tyndall effect, not filterable.
- Everyday importance: sugar and salt production, ORS, blood donation and tests, water and sewage treatment, waste segregation and recycling (eg. recovering lithium from old batteries).
- **Safety:** never look directly into a laser beam; copper sulfate is toxic; handle sulfuric acid only under teacher supervision.